changeproof evidence pack: card-billing v1.4 (eu-dora-hybrid, synthetic)

Repository: card-billing
Commits 51bcd91074c0..81ab6076e1e2: 2 changes. Generated 2026-10-07T12:00:00Z by changeproof 0.1.0.

Release decision

Release decision: 1 of 2 changes fail a policy rule.

```
open: 81ab6076e1e2 agent-changes-need-independent-approval: noreply@anthropic.com took part in this
    change and no independent approver is recorded
    (git-commit/81ab6076e1e2b55456ff7818eee2fd75bfa5fdf0:8)
```

How to check this pack

Everything in this report is read from the signed attestations in the attestations folder, and nothing else. Run: changeproof pack-verify <pack folder> --trust <public key> ... It verifies every signature offline and rebuilds this report and the OSCAL files from the attestations; each must match the digest in pack.dsse.json byte for byte.
Behavioral equivalence means the characterization tests found no difference. It is evidence from tests, not a proof for all inputs. Programs it could not test are listed.

Crypto profile

Crypto profile: hybrid

```
evidence signing: ml-dsa-87, ecdsa-p384; release signing: lms-sha256-192; hash: sha-384
this pack is signed with ml-dsa-87 (key fac2216cb84a03aa)
this pack is signed with ecdsa-p384 (key 90729c14b7040b4e)
release signature: lms-sha256-192 (key 2b244cf4b9bfbac8)
```

Change 1 of 2: commit ed5641a66abe

Who and why

```
author: priya.raman@billing.example (git author,
    git-commit/ed5641a66abe2d693afbff43f5bb5732d5c2bd96:3)
committer: priya.raman@billing.example (git committer,
    git-commit/ed5641a66abe2d693afbff43f5bb5732d5c2bd96:4)
approver: dana.okafor@billing.example (Approved-by trailer,
    git-commit/ed5641a66abe2d693afbff43f5bb5732d5c2bd96:9)
agent: feebot@tools.example (Assisted-by trailer,
    git-commit/ed5641a66abe2d693afbff43f5bb5732d5c2bd96:8)
ticket: LOAN-2 (commit subject, git-commit/ed5641a66abe2d693afbff43f5bb5732d5c2bd96:6)
```

What it touches: 2 entities changed, 2 impacted (2 definite, 0 probable, 0 possible), 0 gaps reported. Touches crypto: no.

```
src/batch/FEECALC.cbl:15-41  paragraph MAIN-PARA (modified)
src/batch/FEECALC.cbl:35  flow COMPUTE (modified)
src/batch/FEECALC.cbl:11  data LK-FEE (definite)
src/batch/FEECALC.cbl:1-41  program FEECALC (definite)
relied on by statement-print through batch-fees
```

Inside the impact set: FEECALC 14 tests, 0 different: no behaviour change found.
Outside the impact set: equivalent. 37 tests (outside-impact-set): 37 same, 0 different, 0 errors.
Linked to this change's impact attestation by its digest.
Policy rules (OPA 1.21.1): all hold.

```
no-new-quantum-vulnerable-crypto: pass
agent-changes-need-independent-approval: pass
equivalence-required-outside-impact-set: pass
```

Change 2 of 2: commit 81ab6076e1e2

Who and why

```
author: priya.raman@billing.example (git author,
    git-commit/81ab6076e1e2b55456ff7818eee2fd75bfa5fdf0:3)
committer: priya.raman@billing.example (git committer,
    git-commit/81ab6076e1e2b55456ff7818eee2fd75bfa5fdf0:4)
agent: noreply@anthropic.com (Co-authored-by trailer,
    git-commit/81ab6076e1e2b55456ff7818eee2fd75bfa5fdf0:8)
ticket: LOAN-3 (commit subject, git-commit/81ab6076e1e2b55456ff7818eee2fd75bfa5fdf0:6)
```

What it touches: 2 entities changed, 2 impacted (2 definite, 0 probable, 0 possible), 0 gaps reported. Touches crypto: no.

```
src/batch/FEECALC.cbl:15-41  paragraph MAIN-PARA (modified)
src/batch/FEECALC.cbl:24  flow COMPUTE (modified)
src/batch/FEECALC.cbl:11  data LK-FEE (definite)
src/batch/FEECALC.cbl:1-41  program FEECALC (definite)
relied on by statement-print through batch-fees
```

Inside the impact set: FEECALC 14 tests, 5 different: behaviour this change altered, to check against its ticket.

```
different: FEECALC test FEECALC-05 (src/batch/FEECALC.cbl:1-41)
different: FEECALC test FEECALC-06 (src/batch/FEECALC.cbl:1-41)
different: FEECALC test FEECALC-08 (src/batch/FEECALC.cbl:1-41)
different: FEECALC test FEECALC-10 (src/batch/FEECALC.cbl:1-41)
different: FEECALC test FEECALC-13 (src/batch/FEECALC.cbl:1-41)
```

Outside the impact set: equivalent. 37 tests (outside-impact-set): 37 same, 0 different, 0 errors.

Linked to this change's impact attestation by its digest.

Policy rules (OPA 1.21.1): a rule fails.

```
no-new-quantum-vulnerable-crypto: pass
agent-changes-need-independent-approval: fail
  deny: noreply@anthropic.com took part in this change and no independent approver is recorded
    (git-commit/81ab6076e1e2b55456ff7818eee2fd75bfa5fdf0:8)
equivalence-required-outside-impact-set: pass
```

Controls this evidence answers

Mapped by the engine's framework table. A control number marked unverified has not been read in its source yet. Planned evidence is not in this pack.

```
DORA Art. 9(4)(e) (source unverified): ICT change management procedures (cited by RTS 2024/1774 Art.
    17(1))
  change record: who, what, when, where and how for every commit
  impact predicate: changed and impacted entities, each with provenance
DORA Art. 28(3) (source unverified): register of information on ICT third-party contractual
    arrangements
  impact predicate: reliant_systems, from the component config
DORA RTS 2024/1774 Art. 17(1)(a) (source checked 2026-10-01): ICT change management: verification of
    whether ICT security requirements have been met
  impact predicate: changed and impacted entities, each with provenance
  impact predicate: touches_crypto when a change touches a crypto call
  behavioral-equivalence attestation: characterization tests from the base replayed at the head, for
    COBOL linkage subprograms outside the impact set
DORA RTS 2024/1774 Art. 17(1)(b) (source checked 2026-10-01): ICT change management: independence of
    the functions that approve changes from those that request and implement them
  change record: requester, implementer and approvers kept apart; self-approval flagged
  policy gate: high-criticality-needs-two-approvers
DORA RTS 2024/1774 Art. 17(1)(d) (source checked 2026-10-01): ICT change management: documentation
    of purpose and scope, timeline and expected outcomes
  change record: ticket and change request copied from the commit, never guessed
  change record: who, what, when, where and how for every commit
  impact predicate: changed and impacted entities, each with provenance
DORA RTS 2024/1774 Art. 17(1)(f)-(g) (source checked 2026-10-01): ICT change management: emergency
    changes, and their review and approval after implementation
  change record: emergency flag from the Change-Type trailer
  change record: requester, implementer and approvers kept apart; self-approval flagged
DORA RTS 2024/1774 Art. 17(1)(h) (source checked 2026-10-01): ICT change management: potential
    impact of a change on existing ICT security measures
  impact predicate: changed and impacted entities, each with provenance
  impact predicate: confidence tier and dependency path per impacted entity
  impact predicate: touches_crypto when a change touches a crypto call
```

  policy gate: no-new-quantum-vulnerable-crypto blocks a change that adds RSA, ECC, DSA or DH
DORA RTS 2024/1774 Art. 16(2)-(3) (source checked 2026-10-01): ICT systems acquisition, development and maintenance: testing and approval before use and after maintenance; source code reviews
  impact predicate: changed and impacted entities, each with provenance
  test subset selected to cover the impact set
  behavioral-equivalence attestation: characterization tests from the base replayed at the head, for COBOL linkage subprograms outside the impact set
ECB ITRQ Q23c-e (source checked 2026-09-27): changes that caused issues, by cause: unexpected interdependencies between applications
  impact predicate: changed and impacted entities, each with provenance
  impact predicate: confidence tier and dependency path per impacted entity
  graph: edges and shared resources marked where one component reaches another
ECB ITRQ Q23c-e (source checked 2026-09-27): changes that caused issues, by cause: inadequate test coverage
  impact predicate: changed and impacted entities, each with provenance
  test subset selected to cover the impact set
  behavioral-equivalence attestation: characterization tests from the base replayed at the head, for COBOL linkage subprograms outside the impact set
ECB ITRQ Q23a (source checked 2026-09-27): emergency changes
  change record: emergency flag from the Change-Type trailer
  change record: requester, implementer and approvers kept apart; self-approval flagged
EU PQC roadmap: no controls mapped (a roadmap with dates, not controls; mapped with the crypto inventory (planned, Week 16))
GDPR: no controls mapped (applies to data egress (ADR 003), not to change evidence)

Attestations

attestations/01-ed5641a66abe-impact.dsse.json 1fde3093861685e01437a160
attestations/01-ed5641a66abe-policy-decision.dsse.json da9063df22170f9ab1cf75b1
attestations/01-ed5641a66abe-behavioral-equivalence.dsse.json ddba7447bcde8d94235d9ec0
attestations/01-ed5641a66abe-full-suite.dsse.json 100ccadd00c4f01ac1a6cfc7
attestations/02-81ab6076e1e2-impact.dsse.json afa4330b38e5fb4864b1b8c6
attestations/02-81ab6076e1e2-policy-decision.dsse.json 9e83605081fd838c33d1290e
attestations/02-81ab6076e1e2-behavioral-equivalence.dsse.json 93ad124f9a9c868903122d11
attestations/02-81ab6076e1e2-full-suite.dsse.json fc71b10f4a9b2ea41bfbefb4